

Industrial Internship Report on
"Crop and Weed detection using YOLO"
&
"Prediction of Agricultural Crop Production in India"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My projects were Crop and weed detection & Prediction of Agricultural Crop Production in India.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

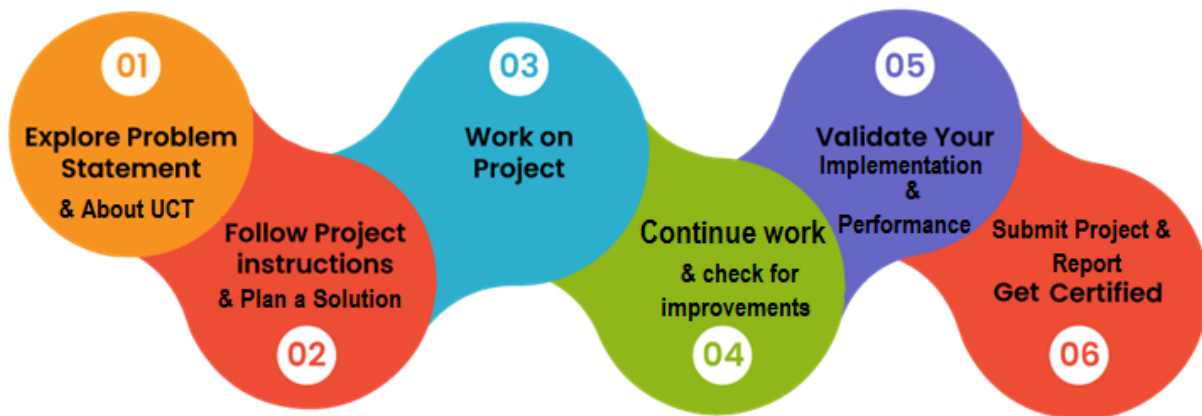
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



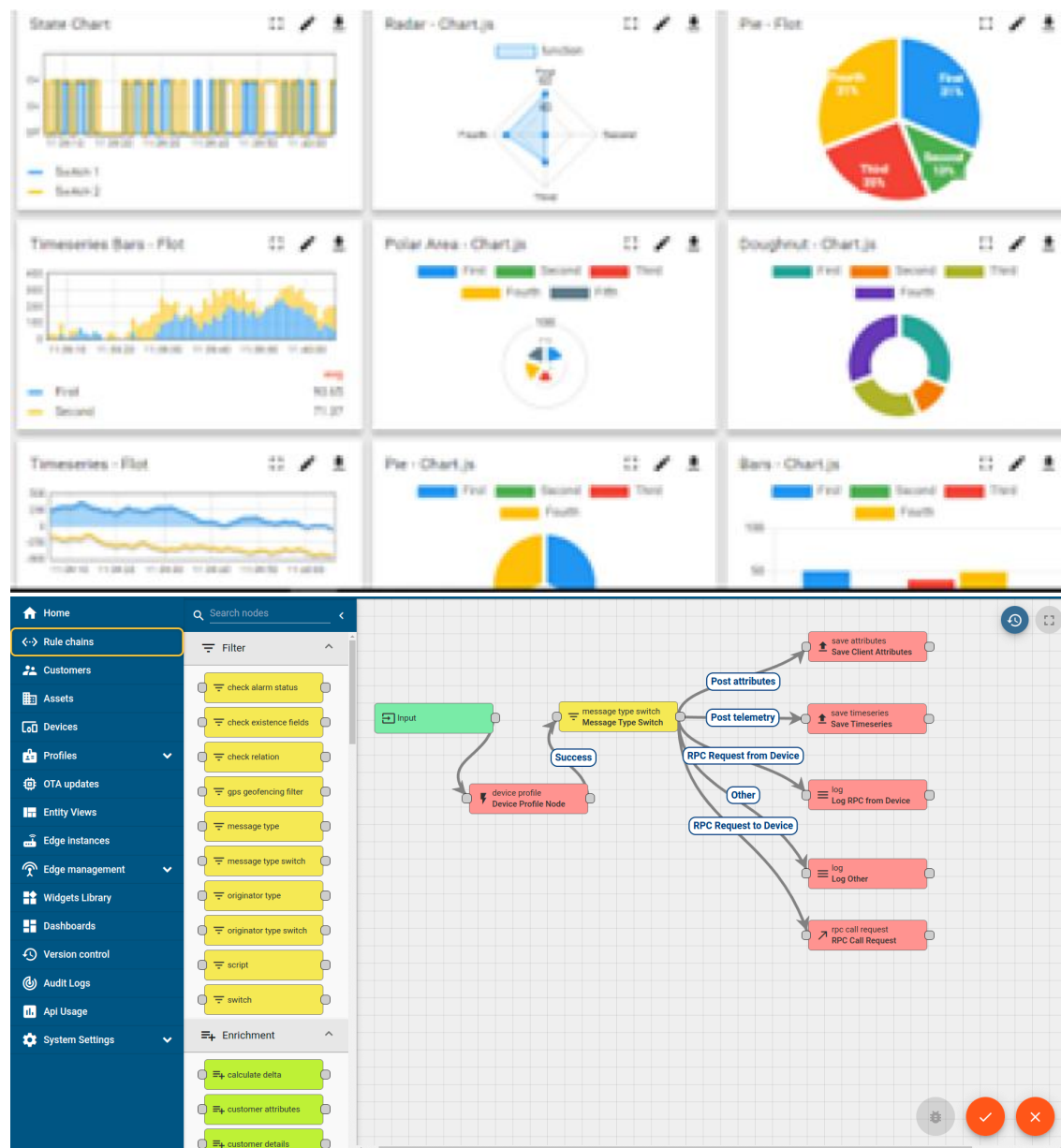
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



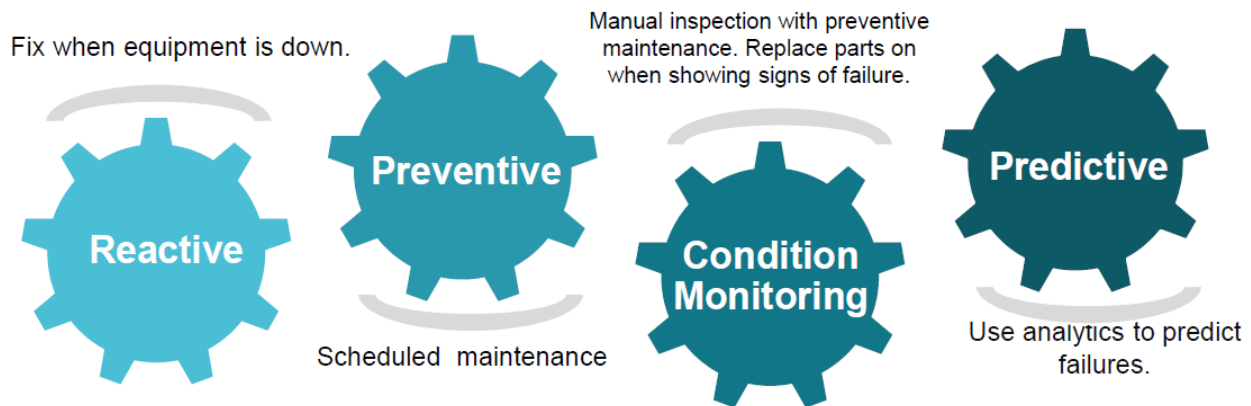


iii. based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

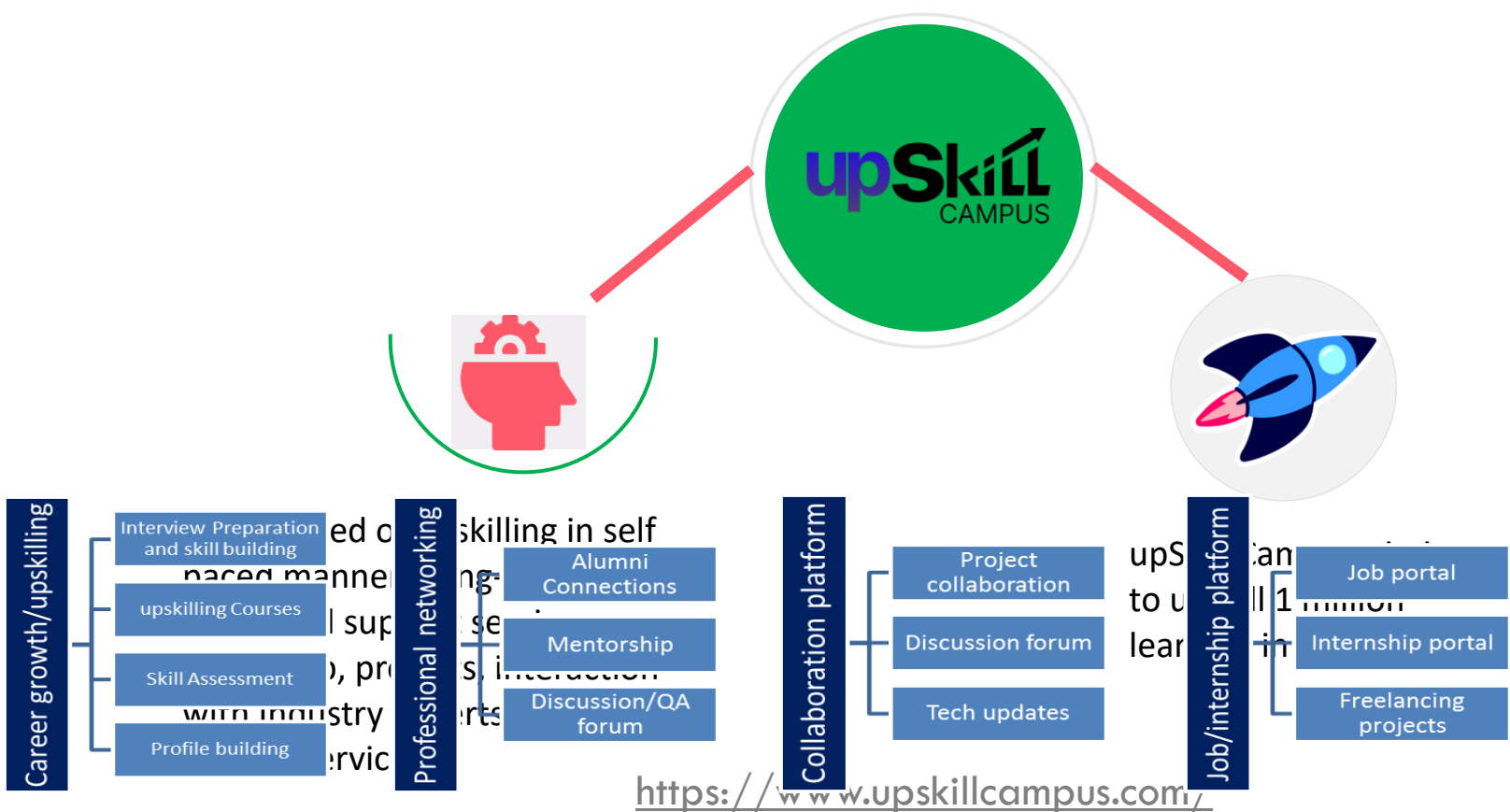
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.

▪ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Python Documentation

[2] Kaggle

[3] Ultralytics Yolo Documentation

2.6 Glossary

Terms	Acronym
ML	Machine Learning
EDA	Exploratory Data Analysis
AI	Artificial Intelligence
YOLO	You Only Look Once
CV	Computer Vision

3 Problem Statement

Weeds are unwanted plants that grow alongside crops and compete for essential resources such as water, nutrients, sunlight, and space. This competition reduces crop growth, lowers agricultural productivity, and increases farming costs. Traditional weed control methods involve manually removing weeds or spraying pesticides across the entire field. Manual weeding is labor-intensive and time-consuming, while indiscriminate pesticide spraying increases chemical usage, harms crops, contaminates soil and water, and poses health risks to humans and the environment.

There is a need for an intelligent and automated system that can accurately distinguish crops from weeds in real time and identify the exact location of weeds. Such a system would enable precision agriculture by allowing pesticides to be applied only where weeds are present, reducing chemical consumption, protecting crops, lowering production costs, and improving crop yield.

This project proposes a **YOLO (You Only Look Once) based Crop and Weed Detection System** that uses computer vision and deep learning to detect and classify crops and weeds from field images. The detected weed locations can later be integrated with an automated spraying mechanism to perform targeted pesticide application, making farming more efficient, economical, and environmentally sustainable.

4 Existing and Proposed solution

Existing Solution:

Currently, weed management in agriculture is primarily carried out through manual weeding and conventional pesticide spraying. Manual weeding involves removing weeds by hand or using agricultural tools, which is effective for small farms but requires significant time, labor, and cost. Another common practice is blanket herbicide spraying, where pesticides are applied uniformly across the entire field without distinguishing between crops and weeds. Although this method controls weed growth, it results in excessive chemical usage, increased production costs, environmental pollution, and potential damage to healthy crops. Some advanced systems use traditional image processing techniques such as color segmentation and edge detection for weed detection, but their performance is often affected by varying lighting conditions, complex backgrounds, and different weed species, making them less reliable for real-world agricultural applications.

Proposed Solution:

The proposed system utilizes the **YOLO (You Only Look Once)** deep learning algorithm to automatically detect and classify crops and weeds from field images in real time. A dataset containing labeled images of crops and weeds is used to train the YOLO model so that it can accurately identify each object and draw bounding boxes around them. During operation, images captured by a camera or drone are processed by the trained model to detect the exact locations of weeds while distinguishing them from crops. The detected weed coordinates can then be used to activate a precision spraying mechanism that applies herbicide only to the identified weeds. This targeted approach significantly reduces pesticide consumption, minimizes crop damage, lowers labor costs, and supports environmentally sustainable farming practices. The proposed solution offers higher detection accuracy, faster processing, and better adaptability to real-field conditions, making it an efficient and practical approach for modern precision agriculture.

4.1 Code submission (Github link)

<https://github.com/dinkar0777/upskillCampus/blob/main/CropAndWeedDetectionUsingYolo.py>

4.2 Report submission (Github link) :

https://github.com/dinkar0777/upskillCampus/blob/main/CropAndWeedDetectionUsingYolo_Dinkar_USC_UCT.pdf

5 Proposed Design/ Model

Given more details about design flow of your solution.

5.1 High Level Diagram (if applicable)

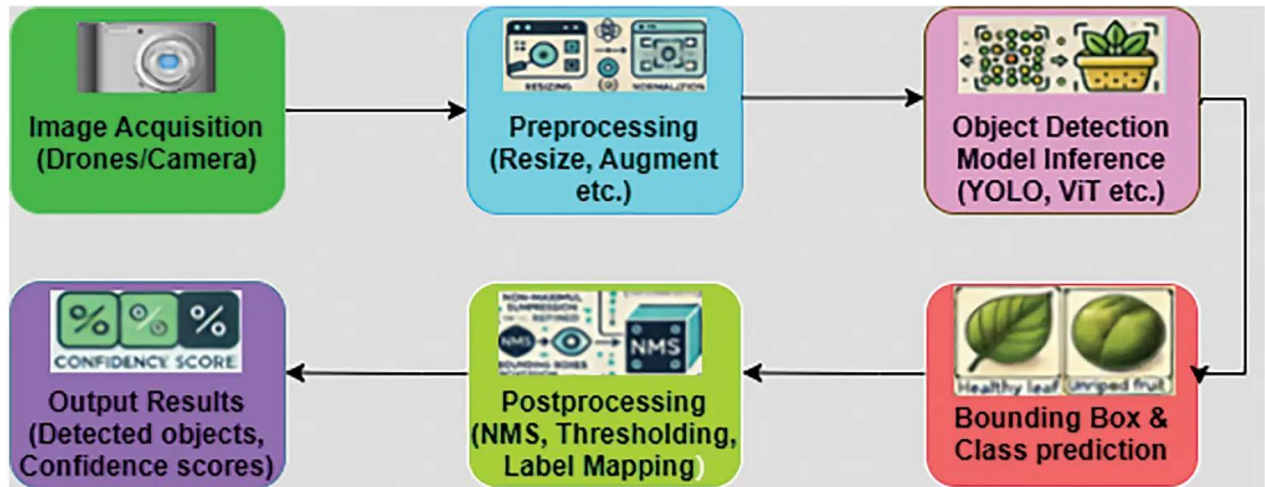
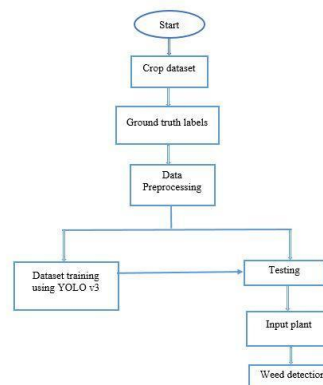


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

- **Programming Language** : Python
- **Development environment** : VS code, Google Colab
- **Libraries** : Pandas, Numpy, Matplotlib, seaborn, scikit-Learn
- **Input** : Agricultural YOLO Dataset

6 Performance Test

6.1 Test Plan/ Test Cases

- **TC-01**: Verify successful upload of valid field images.
- **TC-02**: Check whether the system correctly detects crops.
- **TC-03**: Check whether the system correctly detects weeds.
- **TC-04**: Verify detection when both crops and weeds are present in the same image.
- **TC-05**: Test the system with an invalid or unsupported image file.
- **TC-06**: Verify the system's response when no crops or weeds are present in the image.
- **TC-07**: Ensure that the processed image with detection results is saved successfully.
- **TC-08**: Evaluate the model's detection accuracy and performance using multiple test images.

6.2 Test Procedure

1. Load the trained YOLO model (`best.pt`).
2. Upload or select a test image from the dataset.
3. Preprocess the input image.
4. Run the image through the YOLO detection model.
5. Detect and classify crops and weeds.
6. Display bounding boxes and confidence scores.
7. Save the processed output image.
8. Compare the output with the expected detection results.
9. Record the detection accuracy and system performance.
10. Repeat the process for multiple test images to validate the system.

6.3 Performance Outcome

- Successfully detected both crops and weeds in field images.
- Generated accurate bounding boxes with confidence scores.
- Performed real-time object detection with low processing time.
- Reduced false detections compared to traditional image processing methods.
- Saved processed images successfully after detection.
- Handled invalid or unsupported input images without system failure.
- Demonstrated reliable and consistent performance across multiple test images.
- Suitable for future integration with automated precision spraying systems.

7 My learnings

- . Understood the fundamentals of computer vision and object detection.
- Learned to prepare and annotate datasets for YOLO training.
- Trained and tested a YOLO model for crop and weed detection.
- Gained practical experience with Python, OpenCV, and the Ultralytics YOLO framework.
- Developed a Flask-based application for running the trained model.
- Learned to evaluate model performance and interpret detection results.
- Improved debugging and troubleshooting skills while handling project challenges.
- Understood the real-world applications of AI and deep learning in precision agriculture.
- Enhanced project documentation, testing, and software development skills.

8 Future work scope

- Integrate the system with **automatic pesticide spraying mechanisms** for selective weed control.
- Deploy the model on **drones and autonomous agricultural robots** for large-scale field monitoring.
- Improve detection accuracy by training on **larger and more diverse datasets**.
- Extend the system to identify **multiple crop and weed species**.
- Develop a **mobile application** for real-time detection and monitoring.
- Integrate **IoT sensors** for smart farming and precision agriculture.
- Optimize the model to run efficiently on **edge devices** such as Raspberry Pi and NVIDIA Jetson.
- Implement **cloud-based monitoring and data storage** for remote access and analytics.
- Enhance the system to work effectively under varying weather and lighting conditions.
- Add **yield estimation and crop health monitoring** features to provide a complete smart agriculture solution.